

Chapter 3

Involutions

In this chapter, we define the standard involution on a quaternion algebra. In this way, we characterize division quaternion algebras as noncommutative division rings equipped with a standard involution.

3.1 ► Conjugation

The quaternion conjugation map (2.4.6) defined on the Hamiltonians $\mathbb{H}$ arises naturally from the notion of real and pure (imaginary) parts, as defined by Hamilton. This involution has a natural generalization to a quaternion algebra $B = (a, b \mid F)$ over a field F with $\text{char } F \neq 2$: we define

$$\begin{aligned} \bar{} : B &\rightarrow B \\ \alpha = t + xi + yj + zij &\mapsto \bar{\alpha} = t - (xi + yj + zij) \end{aligned}$$

Multiplying out, we then verify that

$$\alpha\bar{\alpha} = \bar{\alpha}\alpha = t^2 - ax^2 - by^2 + abz^2 \in F.$$

The way in which the cross terms cancel, because the basis elements i, j, ij skew commute, is a calculation that never fails to enchant!

But this definition seems to depend on a basis: it is not intrinsically defined. What properties characterize it? Is it unique? We are looking for a good definition of conjugation $\bar{} : B \rightarrow B$ on an F -algebra B : we will call such a map a *standard involution*.

The involutions we consider should have basic linearity properties: they are F -linear (with $\bar{1} = 1$, so they act as the identity on F) and have order 2 as an F -linear map. An involution should also respect the multiplication structure on B , but we should not require that it be an F -algebra isomorphism: instead, like the inverse map (or transpose map) reverses order of multiplication, we ask that $\overline{\alpha\beta} = \bar{\beta}\bar{\alpha}$ for all $\alpha \in B$. Finally, we want the standard involution to give rise to a trace and norm (a measure of size), which is to say, we want $\alpha + \bar{\alpha} \in F$ and $\alpha\bar{\alpha} = \bar{\alpha}\alpha \in F$ for all $\alpha \in B$. The precise definition is given in Definition 3.2.1, and the defining properties are rigid: if an algebra B has a standard involution, then it is necessarily unique (Corollary 3.4.4).

The existence of a standard involution on B implies that every element of B satisfies a quadratic equation: by direct substitution, we see that $\alpha \in B$ is a root of the polynomial $x^2 - tx + n \in F[x]$ where $t := \alpha + \bar{\alpha}$ and $n := \alpha\bar{\alpha} = \bar{\alpha}\alpha$, since then

$$\alpha^2 - (\alpha + \bar{\alpha})\alpha + \alpha\bar{\alpha} = 0$$

identically. Accordingly, we define the **reduced trace** $\text{trd}: B \rightarrow F$ by $\text{trd}(\alpha) = \alpha + \bar{\alpha}$ and **reduced norm** $\text{nrd}: B \rightarrow F$ by $\text{nrd}(\alpha) = \alpha\bar{\alpha}$. We observe that trd is F -linear and nrd is multiplicative on $B^\times$.

Motivated by this setting, we say that B has **degree 2** if every element $\alpha \in B$ satisfies a (monic) polynomial in $F[x]$ of degree 2 and, to avoid trivialities, that $B \neq F$ (or equivalently, at least one element of B satisfies no polynomial of degree 1). The final result of this section is the following theorem (see Theorem 3.5.1).

Theorem 3.1.1. *Let B be a division F -algebra of degree 2 over a field F with $\text{char } F \neq 2$. Then either $B = K$ is a quadratic field extension of F or B is a division quaternion algebra over F .*

As a consequence, division quaternion algebras are characterized as noncommutative division algebras with a standard involution, when $\text{char } F \neq 2$.

3.2 Involutions

Throughout this chapter, let B be an F -algebra. For the moment, we allow F to be of arbitrary characteristic. We begin by defining involutions on B .

Definition 3.2.1. An **involution** $\bar{}: B \rightarrow B$ is an F -linear map which satisfies:

- (i) $\bar{1} = 1$;
- (ii) $\bar{\bar{\alpha}} = \alpha$ for all $\alpha \in B$; and
- (iii) $\overline{\alpha\beta} = \bar{\beta}\bar{\alpha}$ for all $\alpha, \beta \in B$ (the map $\bar{}$ is an **anti-automorphism**).

3.2.2. We define the **opposite algebra** of B by letting $B^{\text{op}} = B$ as F -vector spaces but with multiplication $\alpha \cdot_{\text{op}} \beta = \beta \cdot \alpha$ for $\alpha, \beta \in B$.

One can then equivalently define an involution to be an F -algebra isomorphism $B \xrightarrow{\sim} B^{\text{op}}$ whose underlying F -linear map has order at most 2.

Remark 3.2.3. What we have defined to be an involution is known in other contexts as an **involution of the first kind**. An **involution of the second kind** is a map which acts nontrivially when restricted to F , and hence is not F -linear; although these involutions are interesting in other contexts, they will not figure in our discussion (and anyway one can consider such an algebra over the fixed field of the involution).

Definition 3.2.4. An involution $\bar{}$ is **standard** if $\alpha\bar{\alpha} \in F$ for all $\alpha \in B$.

Remark 3.2.5. Standard involutions go by many other names. The terminology *standard* is employed because conjugation on a quaternion algebra is the “standard” example of such an involution. Other authors call the standard involution the *main*

involution for quaternion algebras, but then find situations where the “main” involution is not standard by our definition. The standard involution is also called *conjugation* on B , but this can be confused with conjugation by an element in $B^\times$. We will see in Corollary 3.4.4 that a standard involution is unique, so it is also called the *canonical involution*; however, there are other circumstances where involutions can be defined canonically that are not standard (like the map induced by $g \mapsto g^{-1}$ on the group ring $F[G]$).

3.2.6. If $\bar{}$ is a standard involution, so that $\alpha\bar{\alpha} \in F$ for all $\alpha \in B$, then

$$(\alpha + 1)(\overline{\alpha + 1}) = (\alpha + 1)(\bar{\alpha} + 1) = \alpha\bar{\alpha} + \alpha + \bar{\alpha} + 1 \in F$$

and hence $\alpha + \bar{\alpha} \in F$ for all $\alpha \in B$ as well; it then also follows that $\alpha\bar{\alpha} = \bar{\alpha}\alpha$, since

$$(\alpha + \bar{\alpha})\alpha = \alpha(\alpha + \bar{\alpha}).$$

Example 3.2.7. The identity map is a standard involution on $B = F$ as an F -algebra. The $\mathbb{R}$ -algebra $\mathbb{C}$ has a standard involution, namely, complex conjugation.

Example 3.2.8. The **adjugate** map

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto A^\dagger = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is a standard involution on $M_2(F)$ since $AA^\dagger = A^\dagger A = ad - bc = \det A \in F$.

Matrix transpose is an involution on $M_n(F)$ but is a standard involution (if and only if $n = 1$).

3.2.9. Suppose $\text{char } F \neq 2$ and let $B = (a, b \mid F)$. Then the map

$$\begin{aligned} \bar{}: B &\rightarrow B \\ \alpha = t + xi + yj + zij &\mapsto \bar{\alpha} = t - xi - yj - zij \end{aligned}$$

defines a standard involution on B and $\bar{\alpha} = 2t - \alpha$. The map is F -linear with $\bar{1} = 1$ and $\overline{\bar{\alpha}} = \alpha$, so properties (i) and (ii) hold. By F -linearity, it is enough to check property (iii) on a basis (Exercise 3.5), and we verify for instance that

$$\overline{ij} = -ij = ji = (-j)(-i) = \bar{j}\bar{i}$$

(see Exercise 3.5). Finally, the involution is standard because

$$(t + xi + yj + zij)(t - xi - yj - zij) = t^2 - ax^2 - by^2 + abz^2 \in F. \quad (3.2.10)$$

Remark 3.2.11. Algebras with involution play an important role in analysis, in particular Banach algebras with involution and C^* -algebras (generally of infinite dimension). A good reference is the text by Dixmier [Dix77] (or the more introductory book by Conway [Con2012]).

3.3 Reduced trace and reduced norm

Let $\bar{} : B \rightarrow B$ be a standard involution on B . We define the **reduced trace** on B by

$$\begin{aligned} \text{trd} : B &\rightarrow F \\ \alpha &\mapsto \alpha + \bar{\alpha} \end{aligned} \quad (3.3.1)$$

and similarly the **reduced norm**

$$\begin{aligned} \text{nrd} : B &\rightarrow F \\ \alpha &\mapsto \alpha \bar{\alpha}. \end{aligned} \quad (3.3.2)$$

Example 3.3.3. For $B = M_2(F)$, equipped with the adjugate map as a standard involution as in Example 3.2.8, the reduced trace is the usual matrix trace and the reduced norm is the determinant.

3.3.4. The reduced trace trd is an F -linear map, since this is true for the standard involution:

$$\text{trd}(\alpha + \beta) = (\alpha + \beta) + \overline{(\alpha + \beta)} = (\alpha + \bar{\alpha}) + (\beta + \bar{\beta}) = \text{trd}(\alpha) + \text{trd}(\beta)$$

for $\alpha, \beta \in B$. The reduced norm nrd is multiplicative, since

$$\text{nrd}(\alpha\beta) = (\alpha\beta)\overline{(\alpha\beta)} = \alpha\beta\bar{\beta}\bar{\alpha} = \alpha\text{nrd}(\beta)\bar{\alpha} = \text{nrd}(\alpha)\text{nrd}(\beta)$$

for all $\alpha, \beta \in B$.

It will be convenient to write

$$\begin{aligned} B^0 &:= \{\alpha \in B : \text{trd}(\alpha) = 0\} \\ B^1 &:= \{\alpha \in B^\times : \text{nrd}(\alpha) = 1\} \end{aligned} \quad (3.3.5)$$

for the F -subspace $B^0 \subseteq B$ of elements of reduced trace 0 and for the subgroup $B^1 \leq B^\times$ of elements of reduced norm 1. We observe that $B^1 \trianglelefteq B^\times$ is normal, by multiplicativity, indeed we have an exact sequence of groups

$$1 \rightarrow B^1 \rightarrow B^\times \xrightarrow{\text{nrd}} F^\times$$

(noting that the reduced norm map need not be surjective).

Lemma 3.3.6. *If B is not the zero ring, then $\alpha \in B$ is a unit (has a two-sided inverse) if and only if $\text{nrd}(\alpha) \neq 0$.*

Proof. Exercise 3.4. □

Lemma 3.3.7. *For all $\alpha, \beta \in B$, we have $\text{trd}(\beta\alpha) = \text{trd}(\alpha\beta)$.*

Proof. We have

$$\mathrm{trd}(\alpha\bar{\beta}) = \mathrm{trd}(\alpha(\mathrm{trd}(\beta) - \beta)) = \mathrm{trd}(\alpha) \mathrm{trd}(\beta) - \mathrm{trd}(\alpha\beta)$$

and so

$$\mathrm{trd}(\alpha\bar{\beta}) = \overline{\mathrm{trd}(\alpha\beta)} = \mathrm{trd}(\beta\bar{\alpha}) = \mathrm{trd}(\alpha) \mathrm{trd}(\beta) - \mathrm{trd}(\beta\alpha)$$

therefore $\mathrm{trd}(\alpha\beta) = \mathrm{trd}(\beta\alpha)$. $\square$

Remark 3.3.8. The maps trd and nrd are called *reduced* for the following reason.

Let A be a finite-dimensional F -algebra, and consider the left regular representation $\lambda: A \hookrightarrow \mathrm{End}_F(A)$ given by left multiplication in A (cf. Proposition 2.3.1, but over F). We then have a (left) trace map $\mathrm{Tr}: A \rightarrow F$ and (left) norm map $\mathrm{Nm}: A \rightarrow F$ given by mapping $\alpha \in A$ to the trace and determinant of the endomorphism $\lambda_\alpha \in \mathrm{End}_F(A)$.

When $A = M_2(F)$, a direct calculation (Exercise 3.12) reveals that

$$\mathrm{Tr}(\alpha) = 2 \mathrm{trd}(\alpha) = 2 \mathrm{tr}(\alpha)$$

(algebra trace, reduced trace, and matrix trace, respectively; there is no difference between left and right), and

$$\mathrm{Nm}(\alpha) = \mathrm{nrd}(\alpha)^2 = \det(\alpha)^2$$

for all $\alpha \in A$, whence the name *reduced*. (To preview the language of chapter 7, this calculation can be efficiently summarized: as a left A -module, A is the sum of two simple A -modules—acting on the columns of a matrix—and the reduced trace and reduced norm represent ‘half’ of this action.)

3.3.9. Since

$$\alpha^2 - (\alpha + \bar{\alpha})\alpha + \alpha\bar{\alpha} = 0 \tag{3.3.10}$$

identically we see that $\alpha \in B$ is a root of the polynomial

$$x^2 - \mathrm{trd}(\alpha)x + \mathrm{nrd}(\alpha) \in F[x] \tag{3.3.11}$$

which we call the **reduced characteristic polynomial** of α . The fact that α satisfies its reduced characteristic polynomial is the **reduced Cayley–Hamilton theorem** for an algebra with standard involution. When $\alpha \notin F$, the reduced characteristic polynomial of α is its minimal polynomial, since if α satisfies a polynomial of degree 1 then $\alpha \in F$.

3.4 Uniqueness and degree

Definition 3.4.1. An F -algebra K with $\dim_F K = 2$ is called a **quadratic algebra**.

Lemma 3.4.2. *Let K be a quadratic F -algebra. Then K is commutative and has a unique standard involution.*

Proof. Let $\alpha \in K \setminus F$. Then $K = F \oplus F\alpha = F[\alpha]$, so in particular K is commutative. Then $\alpha^2 = t\alpha - n$ for unique $t, n \in F$, since $1, \alpha$ is a basis for K .

If $\bar{} : K \rightarrow K$ is any standard involution, then from (3.3.10) and uniqueness we conclude $t = \alpha + \bar{\alpha}$ (and $n = \alpha\bar{\alpha}$), and so any involution must have $\bar{\alpha} = t - \alpha$. On the other hand, there is a unique standard involution $\bar{} : K \rightarrow K$ with $\bar{\alpha} = t - \alpha$: the verification is straightforward (see Exercise 3.5). $\square$

Example 3.4.3. The reduced trace and norm on a quadratic algebra are precisely the usual algebra trace and norm. If $\text{char } F \neq 2$ and $K \supseteq F$ is a quadratic field extension of F , then the standard involution is just the nontrivial element of $\text{Gal}(K | F)$.

Corollary 3.4.4. *If B has a standard involution, then this involution is unique.*

Proof. For any $\alpha \in B \setminus F$, we have from (3.3.10) that $\dim_F F[\alpha] = 2$, so the restriction of the standard involution to $F[\alpha]$ is unique. Therefore the standard involution on B is itself unique. $\square$

We have seen that the equation (3.3.10), implying that if B has a standard involution then every $\alpha \in B$ satisfies a quadratic equation, has figured prominently in the above proofs. To further clarify the relationship between these two notions, we make the following definition.

Definition 3.4.5. The **degree** of B is the smallest $m \in \mathbb{Z}_{\geq 0}$ such that every element $\alpha \in B$ satisfies a monic polynomial $f(x) \in F[x]$ of degree m , if such an integer exists; otherwise, we say B has degree ∞ .

3.4.6. If B has finite dimension $n = \dim_F B < \infty$, then every element of B satisfies a polynomial of degree at most n : if $\alpha \in B$ then the elements $1, \alpha, \dots, \alpha^n$ are linearly dependent over F . Consequently, every finite-dimensional F -algebra has a (well-defined) integer degree, at most n .

Example 3.4.7. By convention, we interpret Definition 3.4.5 as defining the degree of the zero ring to be 0 (since $1 = 0$, the element 0 satisfies the monic polynomial $0x$)—whatever!

If B has degree 1, then $B = F$. If B has a standard involution, then either $B = F$ or B has degree 2 by (3.3.11).

3.5 Quaternion algebras

We are now ready to characterize division algebras of degree 2 when $\text{char } F \neq 2$. (For the case $\text{char } F = 2$, see Chapter 6.)

Theorem 3.5.1. *Suppose $\text{char } F \neq 2$ and let B be a division F -algebra. Then B has degree at most 2 if and only if one of the following hold:*

- (i) $B = F$;
- (ii) $B = K$ is a quadratic field extension of F ; or
- (iii) B is a division quaternion algebra over F .

Proof. From Example 3.4.7, we may suppose that $B \neq F$ and B has degree 2.

Let $i \in B \setminus F$. Then $F[i] = K$ is a (commutative) quadratic F -subalgebra of the division ring B , so $K = F(i)$ is a field. If $K = B$, we are done. Completing the square (since $\text{char } F \neq 2$), we may suppose that $i^2 = a \in F^\times$.

Let $\phi: B \rightarrow B$ be the map given by conjugation by i , i.e., $\phi(\alpha) = i^{-1}\alpha i$. Then ϕ is a K -linear endomorphism of B , thought of as a (left) K -vector space, and ϕ^2 is the identity on B . Therefore ϕ is diagonalizable, and we may decompose $B = B^+ \oplus B^-$ into eigenspaces for ϕ : explicitly, we can always write

$$\alpha = \frac{\alpha + \phi(\alpha)}{2} + \frac{\alpha - \phi(\alpha)}{2} \in B^+ \oplus B^-.$$

We now prove $\dim_K B^+ = 1$. Let $\alpha \in B^+$. Then $L = F(\alpha, i)$ is a field. Since $\text{char } F \neq 2$, and L is a compositum of quadratic extensions of F , the primitive element theorem implies that $L = F(\beta)$ for some $\beta \in L$. But by hypothesis β satisfies a quadratic equation so $\dim_F L = 2$ and hence $L = K$. (For an alternative direct proof of this claim, see Exercise 3.10.)

If $B = B^+ = K$, we are done. So suppose $B^- \neq \{0\}$. We will prove that $\dim_K B^- = 1$. If $0 \neq j \in B^-$ then $i^{-1}ji = -j$, so $i = -j^{-1}ij$ and hence all elements of B^- conjugate i to $-i$. Thus if $0 \neq j_1, j_2 \in B^-$ then j_1j_2 centralizes i and $j_1j_2 \in B^+ = K$. Thus any two nonzero elements of B^- are K -multiples of each other.

Finally, let $j \in B^- \setminus \{0\}$; then $B = B^+ \oplus B^- = K \oplus Kj$ so B has F -basis $1, i, j, ij$ and $ji = -ij$. We claim that $\text{trd}(j) = 0$: indeed, both j and $i^{-1}ji = -j$ satisfy the same reduced characteristic (or minimal) polynomial of degree 2, so $\text{trd}(j) = \text{trd}(-j) = -\text{trd}(j)$ so $\text{trd}(j) = 0$. Thus $j^2 = b \in F^\times$, and B is a quaternion algebra by definition. $\square$

Remark 3.5.2. We need not assume in Theorem 3.5.1 that B is finite-dimensional; somehow, it is a consequence, and every division algebra over F (with $\text{char } F \neq 2$) of degree ≤ 2 is finite-dimensional.

There are algebras of arbitrary (finite or infinite) dimension over F of degree 2: see Exercise 3.15. Also, a boolean ring (see Exercise 3.12) has degree 2 as an $\mathbb{F}_2$ -algebra, and there are such rings of arbitrary dimension over $\mathbb{F}_2$. Such algebras are quite far from being division rings, of course.

Remark 3.5.3. The proof of Theorem 3.5.1 has quite a bit of history, discussed by van Praag [vPr2002] (along with several proofs). See Lam [Lam2005, Theorem III.5.1] for a parallel proof of Theorem 3.5.1. Moore [Moore35, Theorem 14.4] in 1915 studied algebra of matrices over skew fields and in particular the role of involutions, and gives an elementary proof of this theorem (with the assumption $\text{char } F \neq 2$). Dieudonné [Die48, Die53] gave another proof that relies on structure theory for finite-dimensional division algebras.

Corollary 3.5.4. *Let B be a division F -algebra with $\text{char } F \neq 2$. Then B has degree at most 2 if and only if B has a standard involution.*

Proof. In each of the cases (i)–(iii), B has a standard involution; and conversely if B has a standard involution, then B has degree at most 2 (Example 3.4.7). $\square$

Remark 3.5.5. The statement of Corollary 3.5.4 holds more generally—even if B is not necessarily a division ring—as follows. Let B be an F -algebra with $\text{char } F \neq 2$. Then B has a standard involution if and only if B has degree at most 2 [Voi2011b]. However, this is no longer true in characteristic 2 (Exercise 3.12).

Corollary 3.5.6. *Let B be a division F -algebra with $\text{char } F \neq 2$. Then the following are equivalent:*

- (i) B is a quaternion algebra;
- (ii) B is noncommutative and has degree 2; and
- (iii) B is central and has degree 2.

Definition 3.5.7. An F -algebra B is **algebraic** if every $\alpha \in B$ is algebraic over F (i.e., α satisfies a polynomial with coefficients in F).

If B has finite degree (such as when $\dim_F B = n < \infty$), then B is algebraic.

Corollary 3.5.8 (Frobenius). *Let B be an algebraic division algebra over $\mathbb{R}$. Then either $B = \mathbb{R}$ or $B \simeq \mathbb{C}$ or $B \simeq \mathbb{H}$ as $\mathbb{R}$ -algebras.*

Proof. If $\alpha \in B \setminus \mathbb{R}$ then $\mathbb{R}(\alpha) \simeq \mathbb{C}$, so α satisfies a polynomial of degree 2. Thus if $B \neq \mathbb{R}$ then B has degree 2 and either $B \simeq \mathbb{C}$ or B is a division quaternion algebra over $\mathbb{R}$, and hence $B \simeq \mathbb{H}$ by Exercise 2.3(c). $\square$

Example 3.5.9. Division algebras over $\mathbb{R}$ of infinite dimension abound. Transcendental field extensions of $\mathbb{R}$, such as the function field $\mathbb{R}(x)$ or the Laurent series field $\mathbb{R}((x))$, are examples of infinite-dimensional division algebras over $\mathbb{R}$. Also, the free algebra in two (noncommuting) variables is a subring of a division ring B (its “noncommutative ring of fractions”) with center $\mathbb{R}$ and of infinite dimension over $\mathbb{R}$.

Remark 3.5.10. The theorem of Frobenius (Corollary 3.5.8) extends directly to fields F akin to $\mathbb{R}$, as follows. A field is **formally real** if -1 cannot be expressed in F as a sum of squares and **real closed** if F is formally real and has no formally real proper algebraic extension. The real numbers $\mathbb{R}$ and the field of all real algebraic numbers are real closed. A real closed field has characteristic zero, is totally ordered, and contains a square root of each nonnegative element; the field obtained from F by adjoining a root of the irreducible polynomial $x^2 + 1$ is algebraically closed. For these statements, see Rajwade [Raj93, Chapter 15]. Every finite-dimensional division algebra over a real closed field F is either F or $K = F(\sqrt{-1})$ or $B = (-1, -1 \mid F)$.

Remark 3.5.11. Algebras of dimension 3, sitting somehow between quadratic extensions and quaternion algebras, can be characterized in a similar way. If B is an $\mathbb{R}$ -algebra of dimension 3, then either B is commutative or B has a standard involution and is isomorphic to the subring of upper triangular matrices in $M_2(\mathbb{R})$. A similar statement holds for free R -algebras of rank 3 over a (commutative) domain R ; see Levin [Lev2013].

Exercises

Throughout these exercises, let F be a field.

- 1. Let B be an F -algebra and let $\bar{}: B \rightarrow B$ be an F -linear map with $\bar{1} = 1$. Show that $\bar{}$ is an involution if and only if (ii)–(iii) in Definition 3.2.1 hold for a basis of B (as an F -vector space).
- 2. Let $K = F[\alpha]$ be a quadratic F -algebra, with $\alpha^2 = t\alpha - n$ for (unique) $t, n \in F$. Extending linearly, show that there is a unique standard involution $\bar{}: K \rightarrow K$ with the property that $\bar{\alpha} = t - \alpha$, and show that

$$\begin{aligned}\mathrm{trd}(x + y\alpha) &= 2x + ty \\ \mathrm{nrd}(x + y\alpha) &= x^2 + txy + ny^2\end{aligned}$$

for all $x + y\alpha \in F[\alpha]$.

- 3. Verify that the map $\bar{}$ in Example 3.2.9 is a standard involution.
- 4. Determine the standard involution on $K = F \times F$ (with $F \hookrightarrow K$ under the diagonal map).
- 5. Let B be an F -algebra with a standard involution. Show that $0 \neq \alpha \in B$ is a left zerodivisor if and only if α is a right zerodivisor if and only if $\mathrm{nrd}(\alpha) = 0$. In particular, if B is not the zero ring, then $\alpha \in B$ is (left and right) invertible if and only if $\mathrm{nrd}(\alpha) \neq 0$.
- 6. Suppose $\mathrm{char} F \neq 2$, let B be a division quaternion algebra over F , and let $K_1, K_2 \subseteq B$ be quadratic subfields (over F) with $K_1 \cap K_2 = F$. Show that the F -subalgebra of B generated by K_1 and K_2 is equal to B . Conclude that if $1, \alpha, \beta \in B$ are F -linearly independent, then $1, \alpha, \beta, \alpha\beta$ are an F -basis for B . [Hint: use the involution.] By way of counterexample, show that these results need not hold for $B = M_2(F)$.
- 7. Show that $B = M_n(F)$ has a standard involution if and only if $n \leq 2$.
- 8. Let G be a finite group. Show that the F -linear map induced by $g \mapsto g^{-1}$ for $g \in G$ is an involution on the group ring $F[G] = \bigoplus_{g \in G} Fg$. Determine necessary and sufficient conditions for this map to be a standard involution.
- 9. Let B be an F -algebra with a standard involution $\bar{}: B \rightarrow B$. In this exercise, we examine when $\bar{}$ is the identity map.
 - (a) Show that if $\mathrm{char} F \neq 2$, then $x \in B$ satisfies $\bar{x} = x$ if and only if $x \in F$.
 - (b) Suppose that $\dim_F B < \infty$. Show that the identity map is a standard involution on B if and only if (i) $B = F$ or (ii) $\mathrm{char} F = 2$ and B is a quotient of the commutative ring $F[x_1, \dots, x_n]/(x_1^2 - a_1, \dots, x_n^2 - a_n)$ with $a_i \in F$.
- 10. Let $K \supseteq F$ be a field which has degree m as an F -algebra in the sense of Definition 3.4.5. Suppose that $\mathrm{char} F \nmid m$. Show that $[K : F] = m$, i.e., K has degree m in the usual sense. (What happens when $\mathrm{char} F \mid m$?)

11. Let B be an F -algebra with standard involution. Suppose that $\phi: B \xrightarrow{\sim} B$ is an F -algebra automorphism. Show for $\alpha \in B$ that $\overline{\phi(\alpha)} = \phi(\bar{\alpha})$, and therefore that $\text{trd}(\phi(\alpha)) = \text{trd}(\alpha)$ and $\text{nrd}(\phi(\alpha)) = \text{nrd}(\alpha)$. [Hint: consider the map $\alpha \mapsto \phi^{-1}(\phi(\alpha))$.]
12. In this exercise, we explore further the relationship between algebras of degree 2 and those with standard involutions (Remark 3.5.5).
 - (a) Suppose $\text{char } F \neq 2$ and let B be a finite-dimensional F -algebra. Show that B has a standard involution if and only if $\deg_F B \leq 2$.
 - (b) Let $F = \mathbb{F}_2$ and let B be a Boolean ring, a ring such that $x^2 = x$ for all $x \in B$. (Verify that $2 = 0$ in B , so B is an $\mathbb{F}_2$ -algebra.) Prove that B does not have a standard involution unless $B = \mathbb{F}_2$ or $B = \mathbb{F}_2 \times \mathbb{F}_2$, but nevertheless any Boolean ring has degree at most 2.
- 13. Let $B = M_n(F)$, and consider the map $\lambda: B \hookrightarrow \text{End}_F(B)$ by $\alpha \mapsto \lambda_\alpha$ defined by left-multiplication in B . Show that for all $\alpha \in M_n(F)$, the characteristic polynomial of λ_α is the n th power of the usual characteristic polynomial of α . Conclude when $n = 2$ that $\text{tr}(\lambda_\alpha) = 2 \text{tr}(\alpha)$ and $\det(\lambda_\alpha) = \text{nrd}(\alpha)^2$.
14. Considering a slightly different take on the previous exercise: let B be a quaternion algebra over F . Show that the characteristic polynomial of left multiplication by $\alpha \in B$ is equal to that of right multiplication and is the square of the reduced characteristic polynomial. [Hint: if a direct approach is too cumbersome, consider applying the previous exercise and the left regular representation as in 2.3.8.]
15. Let V be an F -vector space and let $t: V \rightarrow F$ be an F -linear map. Let $B = F \oplus V$ and define the binary operation $x \cdot y = t(x)y$ for $x, y \in V$. Show that $\cdot$ induces a multiplication on B , and that the map $x \mapsto \bar{x} = t(x) - x$ for $x \in V$ induces a standard involution on B . [Such an algebra is called an **exceptional algebra** [GrLu2009, Voi2011b].] Conclude that there exists a central F -algebra B with a standard involution in any dimension $n = \dim_F B \geq 3$.
- 16. In this exercise, we mimic the proof of Theorem 3.5.1 to prove that a quaternion algebra over a finite field of odd cardinality is not a division ring, a special case of Wedderburn's little theorem: a finite division ring is a field. Assume for purposes of contradiction that B is a division quaternion algebra over $F = \mathbb{F}_q$ with q odd.
 - (a) Let $i \in B \setminus F$. Show that the centralizer $C_{B^\times}(i) = \{\alpha \in B^\times : i\alpha = \alpha i\}$ of i in $B^\times$ satisfies $C_{B^\times}(i) = F(i)^\times$.
 - (b) Conclude that any noncentral conjugacy class in $B^\times$ has order $q^2 + 1$.
 - (c) Derive a contradiction from the class equation $q^4 - 1 = q - 1 + m(q^2 + 1)$ (where $m \in \mathbb{Z}$).

[For the case q even, see Exercise 6.7; for fun, the eager reader may wish to prove Wedderburn's little theorem for $F = \mathbb{F}_2$ directly.]

17. Derive Euler's identity (1.1.6) that the product of the sum of four squares is again the sum of four squares as follows. Let $F = \mathbb{Q}(x_1, \dots, x_4, y_1, \dots, y_4)$ be a

function field over $\mathbb{Q}$ in 8 variables and consider the quaternion algebra $(-1, -1 \mid F)$. Show (by an explicit universal formula) that if R is any commutative ring and $x, y \in R$ are the sum of four squares in R , then xy is the sum of four squares in R .

18. Suppose $\text{char } F \neq 2$. For an F -algebra B , let

$$V(B) := \{\alpha \in B \setminus F : \alpha^2 \in F\} \cup \{0\}.$$

Let B be a *central* division ring. Show that $V(B)$ is a nonzero vector space if and only if B is a quaternion algebra over F .

19. Let B be an F -algebra with F -basis $e_1, e_2, \dots, e_n$. Let $\bar{} : B \rightarrow B$ be an involution. Show that $\bar{}$ is standard if and only if

$$e_i \bar{e}_i \in F \text{ and } (e_i + e_j) \overline{(e_i + e_j)} \in F \text{ for all } i, j = 1, \dots, n.$$